Credit Risk Analysis & Customer Segmentation

📌 1. Problem Statement

Financial institutions face significant risk while approving loans, as incorrect assessment of applicants can lead to defaults and financial losses.

The objective of this project is to **identify high-risk loan applicants** using financial and demographic data, enabling better and data-driven lending decisions.

📊 2. Data Overview

The dataset contains approximately **48,000 customer records** with the following key attributes:

- Income (AMT_INCOME_TOTAL)
- Loan Amount (AMT_CREDIT)
- EMI (AMT_ANNUITY)
- Age (AGE)
- Employment Duration (EMPLOYMENT_YEARS)
- Family Members (CNT_FAM_MEMBERS)

Additionally, derived financial metrics were created:

- Loan-to-Income Ratio
- EMI Ratio
- Income per person

🛠 3. Approach & Methodology

🧹 Data Cleaning (Excel)

- Converted raw variables into meaningful features:
  - Days → Age, Employment Years
- Created financial ratios:

    - Loan-to-Income Ratio
    - EMI Ratio
- Removed unnecessary and redundant columns

◈ Data Analysis (SQL)

- Performed exploratory data analysis
- Developed a **rule-based risk segmentation model**:
    - High Risk
    - Medium Risk
    - Low Risk
- Analyzed relationships between:
    - Income and risk
    - Loan amount and risk
    - Age and risk
    - Financial Stress Indicators

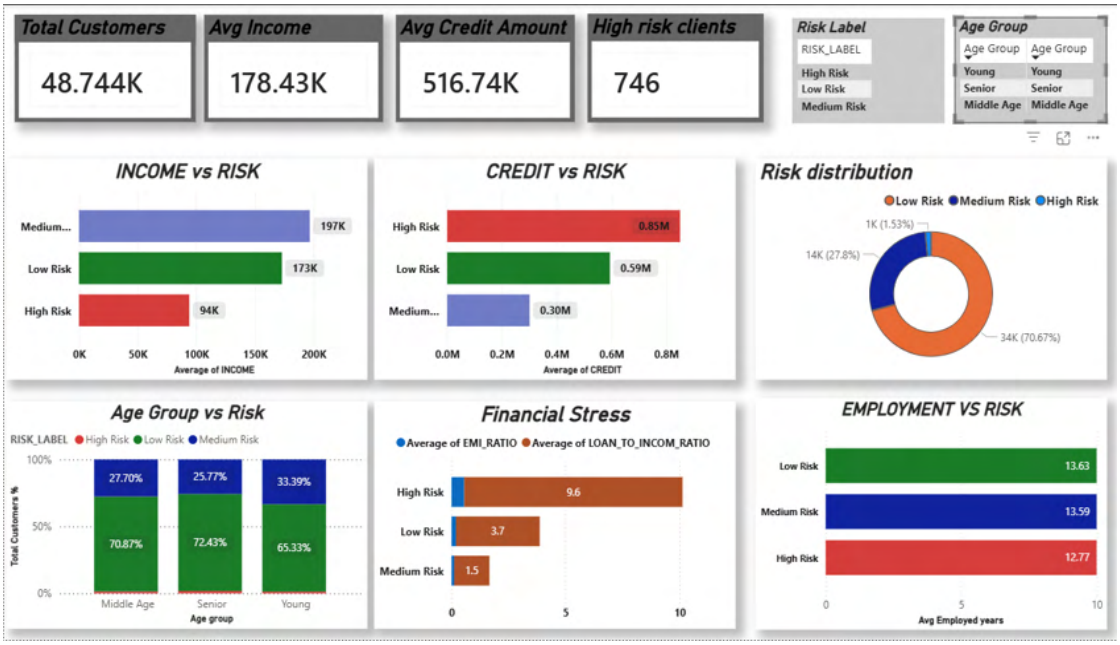
◈ Predictive Modeling (Python)

- Built a **Logistic Regression model** to classify high-risk customers
- Split dataset into training and testing sets (80-20)
- Evaluated model using:
    - Precision
    - Recall
    - F1-score

⚠ Identified and resolved **data leakage** by removing derived features used in target creation.

◈ Data Visualization (Power BI)

- Developed an interactive dashboard including:
    - Risk distribution
    - Income vs risk
    - Credit vs risk
    - Age group analysis
    - Employment trends
    - Financial Stress Indicators



☑ 4. Key Insights

- High-risk customers exhibit significantly **higher loan-to-income ratios**, indicating financial stress
- Lower income groups show a **higher concentration of risk**
- Younger customers tend to have slightly **higher risk compared to older segments**
- Financial burden (EMI and loan ratios) is a **strong indicator of default risk**

🤘 5. Model Performance

- The logistic regression model demonstrated:
  - High precision in identifying high-risk customers
  - Moderate recall (~68%), indicating some risky customers were missed
- Model performance improved after addressing data leakage

💡 6. Business Recommendations

- Implement stricter loan approval criteria for customers with **high loan-to-income ratios**
- Introduce additional verification for **low-income applicants**
- Monitor customers with high EMI burden closely
- Design differentiated strategies for **medium-risk customers** instead of outright rejection

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🚀 7. Conclusion

This project demonstrates how an **end-to-end data analytics approach** can be used to identify high-risk customers and support better financial decision-making.

By integrating Excel, SQL, Python, and Power BI, the project delivers both **analytical insights and practical business recommendations**.